

Quinn and Friends: **Lifeline Link**



Software Requirements Specification

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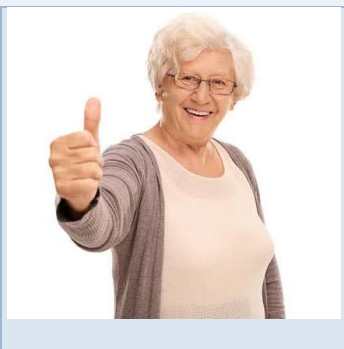
1.Introduction

Around the world, tropical cyclones expose hundreds of millions of people to danger, and natural hazards affect an average of 190 million people annually. With millions of individuals forcibly displaced by climate events each year, the demand for emergency shelter and supplies is always high. The search for localized aid and safe relocation can sometimes be extremely difficult, time-consuming, and taxing during a crisis. That is why we want to create a global website/application that takes some stress off the search for post-hurricane relief. Our website will have an easy-to-use interface that allows users to see all options for humanitarian aid and compare their availability to find the resources that are best for them. It will compare emergency shelter capacity, food distribution locations, medical assistance, power/charging stations, volunteer deployment, and other features to compare and determine the best, most immediate fit for the user.

2.Stakeholders

Dorothy Mae Thompson

The Vulnerable Elder

**AGE**

74

LOCATION

Beaufort County, SC

HOUSEHOLD

Lives alone

OCCUPATION

Retired schoolteacher

STATUS

Displaced — home
flooded

BIO

Dorothy raised her family in the same coastal home for 40 years. When the hurricane hit, rising water forced her to evacuate with only a bag of medications and family photos. She no longer drives long distances and relies on her son, who lives three states away, to help coordinate her care. She is proud and hesitant to ask for help, but genuinely does not know where to turn.

GOALS

Find a safe, accessible place to stay near her community; stay connected with her family and neighbors; get back into a normal routine as quickly as possible.

FRUSTRATIONS

Long shelter registration lines and complicated online forms are hard for her to manage alone. She is anxious about being separated from her medications and unable to reach her regular doctor, and she doesn't want to feel like a burden on overwhelmed relief workers.

CHARACTER

Resilient, but
overwhelmed

3 PROBLEMS THEY NEED SOLVED NOW

- 1.** No stable housing: her home is ruined, and she has nowhere safe to sleep tonight.
- 2.** Mobility and accessibility: she needs transportation and a shelter that can accommodate her limited mobility.
- 3.** Medical Issues: she needs help refilling prescriptions and finding a doctor since her regular pharmacy is closed.

Marcus Reyes

The Protective Parent



AGE

38

LOCATION

Charleston, SC

HOUSEHOLD

Single father of two (ages 6 and 9)

OCCUPATION

Warehouse supervisor

STATUS

Displaced; apartment complex condemned

BIO

As a recently divorced father of two, Marcus works long hours in order to support his children. When the hurricane destroyed his apartment complex, he grabbed his children and the few belongings he could carry out but he lost their important documents and most of their belongings. Marcus is trying to balance finding a shelter, work, and taking care of his kids but is struggling to do so alone.

GOALS

Marcus wishes to keep his family together and in one place. He wants to get his kids back into a routine of some sort, involving food, sleep, and joyful stimulation, no matter how different that may look because of the storm. He also aims to find a shelter as soon as possible in order to return to work and continue providing for his family so he doesn't fall behind once everything returns to normal.

FRUSTRATIONS

Marcus is frustrated because while shelters aim to keep families together many around him are splitting groups by gender or age throughout the day. Additionally, he doesn't have time to navigate multiple aid applications between work shifts, and he

CHARACTER

Determined, stretched
thin

lost his ID and insurance paperwork in the evacuation with no clear way to prove who he is.

3 PROBLEMS THEY NEED SOLVED NOW

1. Family-friendly shelter; he needs a place that keeps him and both kids together, not split apart.
2. Childcare and routine; he needs safe supervision or activities for his kids while he works or handles logistics.
3. Lost documentation; he needs to replace ID, insurance papers, and school records to access aid and enroll his kids in a new school.

Elena Vasquez

The Frontline Responder



AGE

29

LOCATION

Based out of Myrtle Beach, SC

HOUSEHOLD

Lives alone; own apartment damaged

OCCUPATION

EMT / First Responder

STATUS

BIO

Elena has been on shift for over 30 hours straight, transporting patients and coordinating with shelters across the county. Her own apartment took roof damage in the storm, but she hasn't had a moment to deal with it. To do her job well, Elena needs accurate data and information. Eventually, she hopes to have a safe place to sleep once her shift finally ends.

GOALS

Elena needs visibility into shelter capacity and medical resource locations so she can direct people. She also needs a place to rest and recover between shifts due to her home issues. Lastly, she needs to stay in contact with her response team.

FRUSTRATIONS

Conflicting information about which shelters are full and which are still accepting people wastes her time. There's no easy way to see where medical supplies or power stations are available, and she has no time left over to arrange help for her own damaged apartment.

On duty, working
extended shifts

CHARACTER

Selfless, running on empty

3 PROBLEMS THEY NEED SOLVED NOW

- 1.** Proper resource visibility: she needs live, accurate data on shelter capacity and medical resources to do her job effectively.
- 2.** Personal shelter and rest: her own home is damaged, but she has no time to arrange her own housing between shifts.
- 3.** Coordination: she needs a reliable way to communicate with other responders, shelters, and relief organizations in the field.

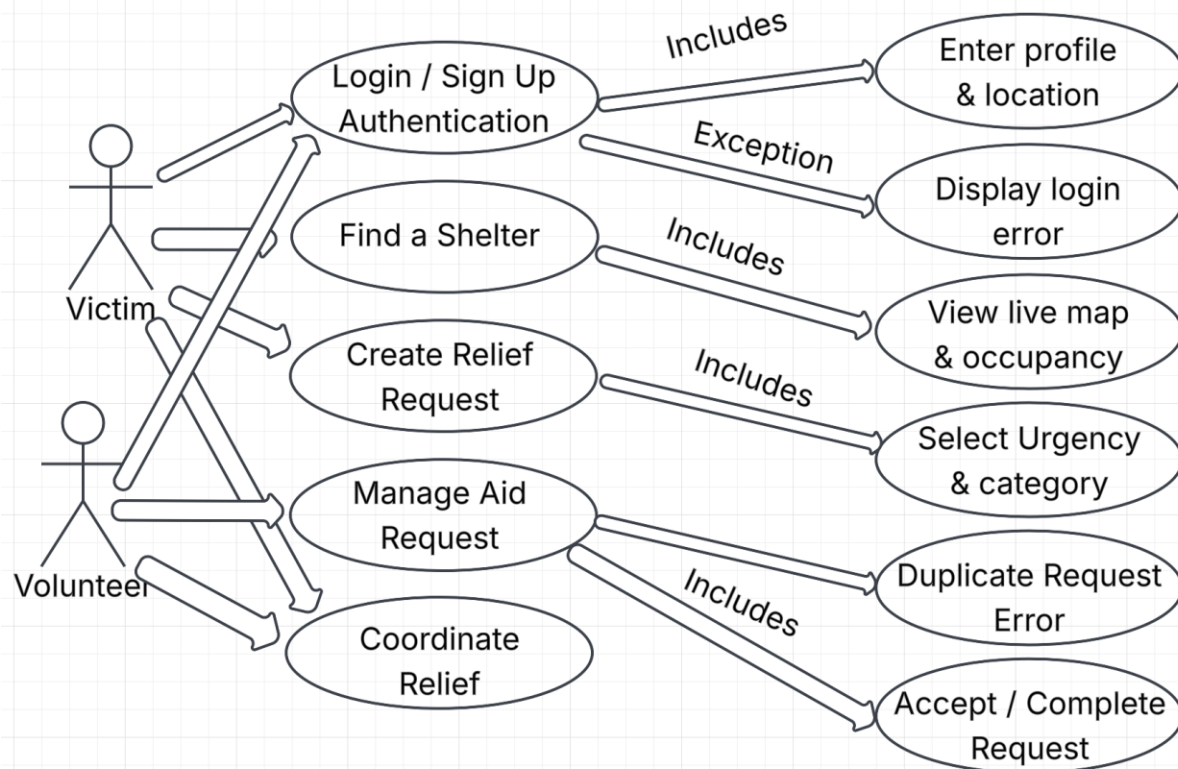
3.Constraints

A 10-week deadline must be met, and a minimum viable product must be reached by Week 8 to allow for testing of the product in the final 2 weeks. Less than \$10,000 is available for the completion of this project including all the prototyping, software licensing, and hardware components. At least 15% of the funds for the project will be set aside for emergency hardware replacements or unexpected shipping fees. To be able to scale the project at a future point in time, no single hardware component can exceed a unit cost of \$500. The cloud infrastructure used in this project must have cloud cost under \$50 a month. A mandatory mid-project design review must be completed and passed during Week 6 before any final software deployment begins.

4.Overall Description

This hurricane relief system needs to run smoothly on pretty much any device people might have during an emergency. It will be built as a cross-platform application so that it works seamlessly across iPhones, Androids, Mac, and Windows computers without needing separate complex downloads. The system will need to connect with external weather APIs to pull real-time storm tracking data and display it to the user. We also plan to interface it with standard GPS hardware on mobile devices to help pinpoint where relief is needed most.

5.Business Use Cases



https://lucid.app/lucidchart/041c5661-800d-40c6-bdbb-397913133d55/edit?viewport_loc=-40%2C-74%2C1151%2C731%2C0_0&invitationId=inv_20244d77-08f6-45f6-b59e-574fca3fab17

6. Functional Requirements

[Full Requirements spreadsheet](#)

7. Non-Functional Requirements

User Requirements

- The application shall utilize a high-contrast color scheme to ensure readability outdoors in direct sunlight or rain.

- The system shall allow users to submit an emergency relief request within a maximum of three taps from the home screen.
- The application shall prioritize battery-efficient operations during poor network connectivity.
- The system shall display error messages containing clear recovery steps without technical jargon.
- The application shall present a mandatory location-sharing consent agreement during account creation.

Admin

- The system backend shall support up to 100,000 concurrent active users without performance degradation.
- The system shall require multi-factor authentication for administrator accounts.
- The system shall require multi-factor authentication for volunteer accounts.
- The system shall maintain a searchable trail of all edits to determine any inconsistencies

Shelter

- The system shall encrypt all medical incident logs

Resources

- The system architecture shall remain modular to allow independent updates of individual features without system-wide downtime.
- The system shall execute automated health checks on third-party integrations every five minutes.

Relief Request

- The system shall mask exact victim coordinates from civilian volunteers until a relief request receives an official acceptance.
- The application shall enforce rate-limiting on request submissions to prevent distributed denial-of-service attacks.
- The user interface shall display a Good Samaritan legal disclaimer for civilian volunteers.

Hurricane

- The application design shall have a calm visual aesthetic.

Geographical Region

- The application shall adapt to distance measurements based on the local settings of the user's device.

8. Definitions and Acronyms

- Define any terms used in the document that may not be obvious to the reader.
- Make sure to list this information in an easy-to-read format.

9. Competitive Analysis

Product	Focus	Pros	Cons
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FEMA App	Government alerts and official recovery resources.	Highly reliable alerts; direct links to federal aid and shelters.	One-way communication; no community reporting features.
Red Cross App	Immediate safety and shelter finding.	"I'm Safe" messaging feature; clear map of official shelters.	Lacks tools for organizing long-term volunteers or supplying relief.
Ushahidi	Crowdsourced crisis mapping and reporting.	Excellent for real-time, community-driven hazard reporting.	User data can be unverified and chaotic without heavy moderation.

Looking at these competitors, our project could stand out by bridging the gap between official information and community action. We should definitely adopt a simple "I'm Safe" status feature like the Red Cross uses, as that is a high-priority need for users. We could also take Ushahidi's interactive community map but improve upon it by adding a verification system so the data doesn't get too cluttered. It is clear that the most important thing is that we offer two-way communication and still have a structured way to get asked and receive help.

10. References

- List any documents referred to in the creation of this requirements document.

11. Appendices

Any relevant information that can assist in understanding the requirements.